| | | |
|---|---|---|
| | | **Data Type** xsd:string |
| **D_337**<br>Time | Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99) | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 6 **max:** 6<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_28**<br>Group Control Number | Assigned number originated and maintained by the sender | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 9 **max:** 9<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_455**<br>Responsible Agency Code | Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 1 **max:** 1<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_455<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_480**<br>Version / Release / Industry Identifier Code | Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 6 **max:** 6<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_480<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |

## 2.5 S_GE — Functional Group Trailer

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **FunctionalGroup** — Functional Group | 1 .. 1 | | | | |
| │ └ **S_GE** — Functional Group Trailer | 1 .. 1 | | | | |
| │ ├ **D_97** — Number of Transaction Sets Included | 1 .. 1 | String | | 1 .. 1 | |
| │ └ **D_28** — Group Control Number | 1 .. 1 | String | | 9 .. 9 | |

### Documentation

**Definition** To indicate the end of a functional group and to provide control information Comments 1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header. Semantic Notes: 1. The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

### Properties

**Identifier** S_GE

**Name** Functional Group Trailer

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Element

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **D_97**<br>Number of Transaction Sets Included | Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 1<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_28**<br>Group Control Number | Assigned number originated and maintained by the sender | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 9 max: 9<br>**External Category** Element<br>**Data Type** xsd:string |

## 2.6 S_IEA — Interchange Control Trailer

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| └ **S_IEA** — Interchange Control Trailer | 1 .. 1 | | | | |
| ├ **D_I16** — Number of Included Functional Groups | 1 .. 1 | String | | 1 .. 1 | |
| └ **D_I12** — Interchange Control Number | 1 .. 1 | String | | 9 .. 9 | |

### Documentation

**Definition** To define the end of an interchange of zero or more functional groups and interchange-related control segments

### Properties

**Identifier** S_IEA

**Name** Interchange Control Trailer

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Element

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **D_I16**<br>Number of Included Functional Groups | A count of the number of functional groups included in an interchange | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 1<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I12**<br>Interchange Control Number | A control number assigned by the interchange sender | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 9 max: 9<br>**External Category** Element<br>**Data Type** xsd:string |

# 3 OVERVIEW MESSAGE

## General Information

| | |
|---|---|
| **Name** | SAP_LBN_ANSI_X12_310 |
| **Direction** | In |
| **Status** | Active |
| **Message Type** | Freight Receipt and Invoice (Ocean) |
| **Type System** | ASC_X12 |
| **Version** | 004010 |
| **External Category** | Transaction Set |

## Documentation

**Definition** This Draft Standard for Trial Use contains the format and establishes the data contents of the Freight Receipt and Invoice (Ocean) Transaction Set (310) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to provide ocean bill of lading information. It is sent by ocean carriers to interested parties and can be used as the receipt for the shipment; to substitute for a paper bill of lading where the parties have agreed that a paper bill of lading is not necessary; to allow shipper or forwarder to verify bill of lading information before an original is printed and released; for information purposes, i.e., as a bill of lading copy; by the carrier to convey manifest information to a terminal operator; and as an invoice for freight.

## Notes

**Example 1**

```
ISA*00*          *00*          *02*10000043       *01*10000042
*250305*2035*U*00401*000000336*0*P*>~
GS*IO*10000043*10000042*20250305*2035*0*X*004010~
ST*310*003360506~
B3**INV202511261009*RTS 893246031*PP**20250305*4817****TNT~
N9*12*5500300965~
N9*CN*0013000000~
N9*06*ERP123~
V1*000ZZ*SL FUTURE~
C3*EUR~
N1*SH*SHIPPER INC~
N3*294-4, AHYEON-DONG, MAPO-GU~
R4*D*UN*USTIW*TACOMA~
K1*ACC NL600000571~
K1*AWB 912477510~
K1*VAT NL857768578B01~
K1*PRD G15N~
LX*1~
L0***KG**********83~
L1**40.89*RC*4817********EXPR_FREIGHT*****1*KG**EUR~
L3*14070*G***111250****41.08*X*756*K~
SE*23*003360506~
GE*1*0~
IEA*1*000000336~
```

### 3.1 Structure

The following table shows the complete structure.

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **310** — Freight Receipt and Invoice (Ocean) | 1 .. 1 | | | | |
| ├ **ST** — Transaction Set Header | 1 .. 1 | | 010 | | |
| │ ├ **143** — Transaction Set Identifier Code | 1 .. 1 | Token | 01 | 3 .. 3 | 143 |
| │ └ **329** — Transaction Set Control Number | 1 .. 1 | String | 02 | 4 .. 9 | |
| ├ **B3** — Beginning Segment for Carrier's Invoice | 1 .. 1 | | 020 | | |
| │ ├ **76** — Invoice Number | 1 .. 1 | String | 02 | 1 .. 22 | |
| │ ├ **145** — Shipment Identification Number | 0 .. 1 | String | 03 | 1 .. 30 | |
| │ ├ **146** — Shipment Method of Payment | 1 .. 1 | Token | 04 | 2 .. 2 | 146 |
| │ ├ **373** — Date | 1 .. 1 | Date | 06 | 8 .. 8 | |
| │ ├ **193** — Net Amount Due | 1 .. 1 | Decimal | 07 | 1 .. 12 | |
| │ └ **140** — Standard Carrier Alpha Code | 1 .. 1 | Token | 11 | 2 .. 4 | |
| ├ **N9** — Reference Identification | 3 .. 15 | | 060 | | |
| │ ├ **128** — Reference Identification Qualifier | 1 .. 1 | Token | 01 | 2 .. 3 | 128 |
| │ ├ **127** — Reference Identification - System Number | 1 .. 1 | String | 02 | 1 .. 30 | |
| │ ├ **127** — Reference Identification - Billing Account | 0 .. 1 | String | 02 | 1 .. 30 | |
| │ ├ **127** — Reference Identification - Carrier's Reference Number (PRO/Invoice) | 0 .. 1 | String | 02 | 1 .. 30 | |
| │ ├ **369** — Free-form Description | 0 .. 1 | String | 03 | 1 .. 45 | |
| │ ├ **373** — Date | 0 .. 1 | Date | 04 | 8 .. 8 | |
| │ ├ **337** — Time | 0 .. 1 | Time | 05 | 4 .. 8 | |
| │ ├ **623** — Time Code | 0 .. 1 | Token | 06 | 2 .. 2 | 623 |
| │ └ **C040** — Reference Identifier | 0 .. 1 | | 07 | | |
| │ ├ **128** — Reference Identification Qualifier | 1 .. 1 | Token | 01 | 2 .. 3 | 128 |
| │ ├ **127** — Reference Identification | 1 .. 1 | String | 02 | 1 .. 30 | |
| │ ├ **128** — Reference Identification Qualifier | 0 .. 1 | Token | 03 | 2 .. 3 | 128 |
| │ ├ **127** — Reference Identification | 0 .. 1 | String | 04 | 1 .. 30 | |
| │ ├ **128** — Reference Identification Qualifier | 0 .. 1 | Token | 05 | 2 .. 3 | 128 |
| │ └ **127** — Reference Identification | 0 .. 1 | String | 06 | 1 .. 30 | |
| ├ **V1** — Vessel Identification | 1 .. 2 | | 070 | | |
| ├ **C3** — Currency | 0 .. 1 | | 110 | | |
| │ └ **100** — Currency Code | 1 .. 1 | Token | 01 | 3 .. 3 | ISO_4217 |
| ├ **N1** — Loop N1 | 1 .. 10 | | 130 | | |
| │ └ **N1** — Name | 1 .. 1 | | 130 | | |
| │ └ **98** — Entity Identifier Code | 1 .. 1 | Token | 01 | 2 .. 3 | 98 |
| ├ **R4** — Loop R4 | 1 .. 20 | | 180 | | |
| │ └ **R4** — Port or Terminal | 1 .. 1 | | 180 | | |
| │ └ **115** — Port or Terminal Function Code | 1 .. 1 | Token | 01 | 1 .. 1 | 115 |
| ├ **K1** — Remarks | 0 .. 12 | | 210 | | |
| │ ├ **61** — Free-Form Message | 1 .. 1 | String | 01 | 1 .. 30 | |
| │ └ **61** — Free-Form Message | 0 .. 1 | String | 02 | 1 .. 30 | |
| ├ **LX** — Loop LX | 1 .. 999 | | 010 | | |
| │ ├ **LX** — Assigned Number | 1 .. 1 | | 010 | | |
| │ │ └ **554** — Assigned Number | 1 .. 1 | Integer | 01 | 1 .. 6 | |
| │ └ **L0** — Loop L0 | 0 .. 120 | | 150 | | |
| │ ├ **L0** — Line Item - Quantity and Weight | 1 .. 1 | | 150 | | |
| │ │ ├ **221** — Billed/Rated-as Qualifier | 0 .. 1 | Token | 03 | 2 .. 2 | 221 |
| │ │ └ **380** — Quantity | 0 .. 1 | Decimal | 13 | 1 .. 15 | |
| │ └ **L1** — Loop L1 | 0 .. 20 | | 170 | | |
| │ └ **L1** — Rate and Charges | 1 .. 1 | | 170 | | |
| │ ├ **60** — Freight Rate | 0 .. 1 | Decimal | 02 | 1 .. 9 | |
| │ ├ **122** — Rate/Value Qualifier | 0 .. 1 | Token | 03 | 2 .. 2 | 122 |
| │ ├ **58** — Charge | 0 .. 1 | Decimal | 04 | 1 .. 12 | |
| │ ├ **150** — Special Charge or Allowance Code | 0 .. 1 | Token | 08 | 3 .. 3 | 150 |
| │ ├ **276** — Special Charge Description | 0 .. 1 | String | 12 | 2 .. 25 | |
| │ ├ **220** — Billed/Rated-as Quantity | 0 .. 1 | Decimal | 17 | 1 .. 11 | |
| │ ├ **221** — Billed/Rated-as Qualifier | 0 .. 1 | Token | 18 | 2 .. 2 | 221 |
| │ ├ **954** — Percent | 0 .. 1 | Decimal | 19 | 1 .. 10 | |
| │ ├ **100** — Currency Code | 0 .. 1 | Token | 20 | 3 .. 3 | ISO_4217 |
| │ └ **610** — Amount | 0 .. 1 | Decimal | 21 | 1 .. 15 | |
| ├ **L3** — Total Weight and Charges | 1 .. 1 | | 010 | | |
| │ ├ **58** — Charge | 0 .. 1 | Decimal | 05 | 1 .. 12 | |
| │ └ **150** — Special Charge or Allowance Code | 0 .. 1 | Token | 08 | 3 .. 3 | 150 |
| └ **SE** — Transaction Set Trailer | 1 .. 1 | | 080 | | |

| | | | | | |
|---|---|---|---|---|---|
| ├ **96** — Number of Included Segments | 1 .. 1 | Integer | 01 | 1 .. 10 | |
| └ **329** — Transaction Set Control Number | 1 .. 1 | String | 02 | 4 .. 9 | |

# 4 DETAILS MESSAGE

## 4.1 310 — Freight Receipt and Invoice (Ocean)

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **310** — Freight Receipt and Invoice (Ocean) | 1 .. 1 | | | | |
| ├ **ST** — Transaction Set Header | 1 .. 1 | | 010 | | |
| ├ **B3** — Beginning Segment for Carrier's Invoice | 1 .. 1 | | 020 | | |
| ├ **N9** — Reference Identification | 3 .. 15 | | 060 | | |
| ├ **V1** — Vessel Identification | 1 .. 2 | | 070 | | |
| ├ **C3** — Currency | 0 .. 1 | | 110 | | |
| ├ **N1** — Loop N1 | 1 .. 10 | | 130 | | |
| ├ **R4** — Loop R4 | 1 .. 20 | | 180 | | |
| ├ **K1** — Remarks | 0 .. 12 | | 210 | | |
| ├ **LX** — Loop LX | 1 .. 999 | | 010 | | |
| ├ **L3** — Total Weight and Charges | 1 .. 1 | | 010 | | |
| └ **SE** — Transaction Set Trailer | 1 .. 1 | | 080 | | |

### Properties

**Identifier** 310

**Name** Freight Receipt and Invoice (Ocean)

**Cardinality** min: 1 max: 1

## 4.2 ST — Transaction Set Header

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **310** — Freight Receipt and Invoice (Ocean) | 1 .. 1 | | | | |
| ├ **ST** — Transaction Set Header | 1 .. 1 | | 010 | | |
| │ ├ **143** — Transaction Set Identifier Code | 1 .. 1 | Token | 01 | 3 .. 3 | 143 |
| │ └ **329** — Transaction Set Control Number | 1 .. 1 | String | 02 | 4 .. 9 | |

### Documentation

**Definition** To indicate the start of a transaction set and to assign a control number

### Properties

**Identifier** ST

**Name** Transaction Set Header

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Segment

**Position** 010

**Level** 2

### Notes

**Usage 1** The transaction set identifier (ST01) used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **143**<br>Transaction Set Identifier Code | Code uniquely identifying a Transaction Set<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** Token<br>**Length** min: 3 max: 3<br>**External Category** Simple Data Element<br>**Data Type** ID<br>**Position** 01<br><br>**Codelist Id** 143<br>**Type System** ASC_X12<br>**Version Mode** Current<br>**Version** 004010 |
| **329**<br>Transaction Set Control Number | Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 4 max: 9<br>**External Category** Simple Data Element<br>**Data Type** AN<br>**Position** 02 |

## 4.3 B3 — Beginning Segment for Carrier's Invoice

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **310** — Freight Receipt and Invoice (Ocean) | 1 .. 1 | | | | |
| ├ **B3** — Beginning Segment for Carrier's Invoice | 1 .. 1 | | 020 | | |
| │ ├ **76** — Invoice Number | 1 .. 1 | String | 02 | 1 .. 22 | |
| │ ├ **145** — Shipment Identification Number | 0 .. 1 | String | 03 | 1 .. 30 | |
| │ ├ **146** — Shipment Method of Payment | 1 .. 1 | Token | 04 | 2 .. 2 | 146 |
| │ ├ **373** — Date | 1 .. 1 | Date | 06 | 8 .. 8 | |
| │ ├ **193** — Net Amount Due | 1 .. 1 | Decimal | 07 | 1 .. 12 | |
| │ └ **140** — Standard Carrier Alpha Code | 1 .. 1 | Token | 11 | 2 .. 4 | |

### Documentation

**Definition** To transmit basic data relating to the carrier's invoice

### Properties

**Identifier** B3

**Name** Beginning Segment for Carrier's Invoice

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Segment

**Position** 020

**Level** 2

### Notes

**Usage 1** B306 is the billing date.

**Usage 2** On collect freight bills the data in B309 will be the actual delivery date. For the shipments invoiced prior to delivery, the B309 will be the estimated date of delivery.

**Usage 3** If B314 is used, B304 will indicate the party or parties responsible for payment of the transportation terms identified in B314.

**Constraint 1** P0910 Paired Multiple - If any are used, all must be used

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **76**<br>Invoice Number | Identifying number assigned by issuer | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 22<br>**External Category** Simple Data Element<br>**Data Type** AN<br>**Position** 02 |
| **145**<br>Shipment Identification Number | Identification number assigned to the shipment by the shipper that uniquely identifies the shipment from origin to ultimate destination and is not subject to modification; (Does not contain blanks or special characters) | **Cardinality** min: 0 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 30<br>**External Category** Simple Data Element<br>**Data Type** AN<br>**Position** 03 |
| **146**<br>Shipment Method of Payment | Code identifying payment terms for transportation charges<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** Token<br>**Length** min: 2 max: 2<br>**External Category** Simple Data Element<br>**Data Type** ID<br>**Position** 04<br><br>**Codelist Id** 146 |

| | | |
|---|---|---|
| | | **Type System** ASC_X12<br>**Version Mode** Current<br>**Version** 004010 |
| **373**<br>Date | Date expressed as CCYYMMDD | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** Date<br>**Length** **min:** 8 **max:** 8<br>**Date Time Format** CCYYMMDD<br>**External Category** Simple Data Element<br>**Data Type** DT<br>**Position** 06 |
| **193**<br>Net Amount Due | Total charges to be paid by the receiver of this transaction set expressed in the standard monetary denomination for the currency specified | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** Decimal<br>**Length** **min:** 1 **max:** 12<br>**Fraction Digits** 2<br>**Total Digits** 12<br>**External Category** Simple Data Element<br>**Data Type** N2<br>**Position** 07 |
| **140**<br>Standard Carrier Alpha Code | Standard Carrier Alpha Code | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** Token<br>**Length** **min:** 2 **max:** 4<br>**External Category** Simple Data Element<br>**Data Type** ID<br>**Position** 11 |